Kinvecta

HOW LARGE LANGUAGE MODELS WORK · EPISODE 6 ·

One step downhill. Land at 19?

Same direction, just a bigger step, and you end up higher than you started. Today we find out why.

12-minute lesson

Companion notebook



MARK

WHY THIS MATTERS

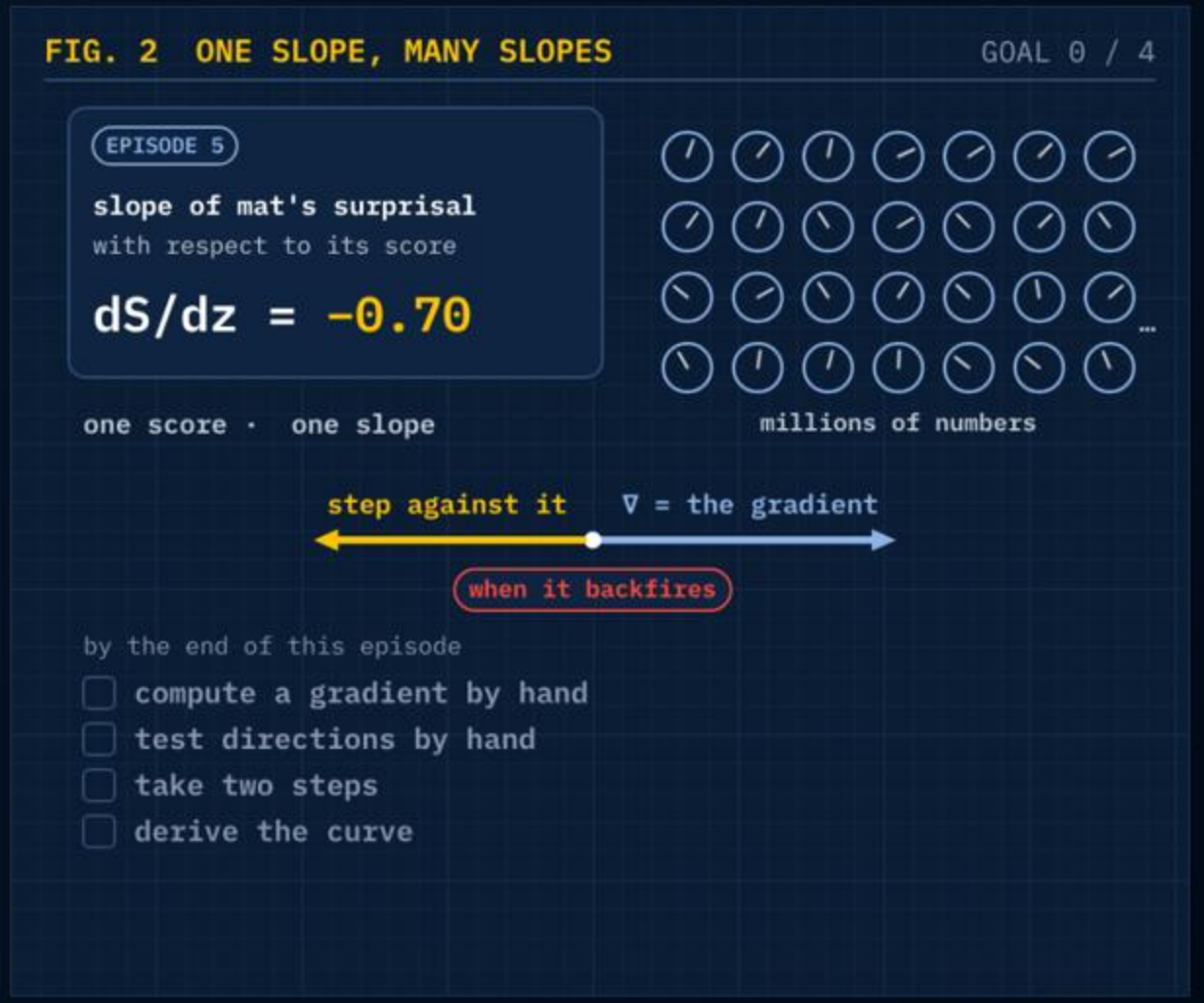
One score, one slope. Now millions.

Episode 5: dS/dz = −0.70 for mat. A model has millions of numbers.

01 Put every slope into one arrow: the gradient

02 Step against it

03 Find exactly when that step backfires



Episode 5 (part 1). Episode 1's probabilities (mat 0.30, floor 0.20, sofa 0.15, other 0.35) are illustrative, not measured.

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EPISODE 6 · CALCULUS FOR LEARNING, PART 2

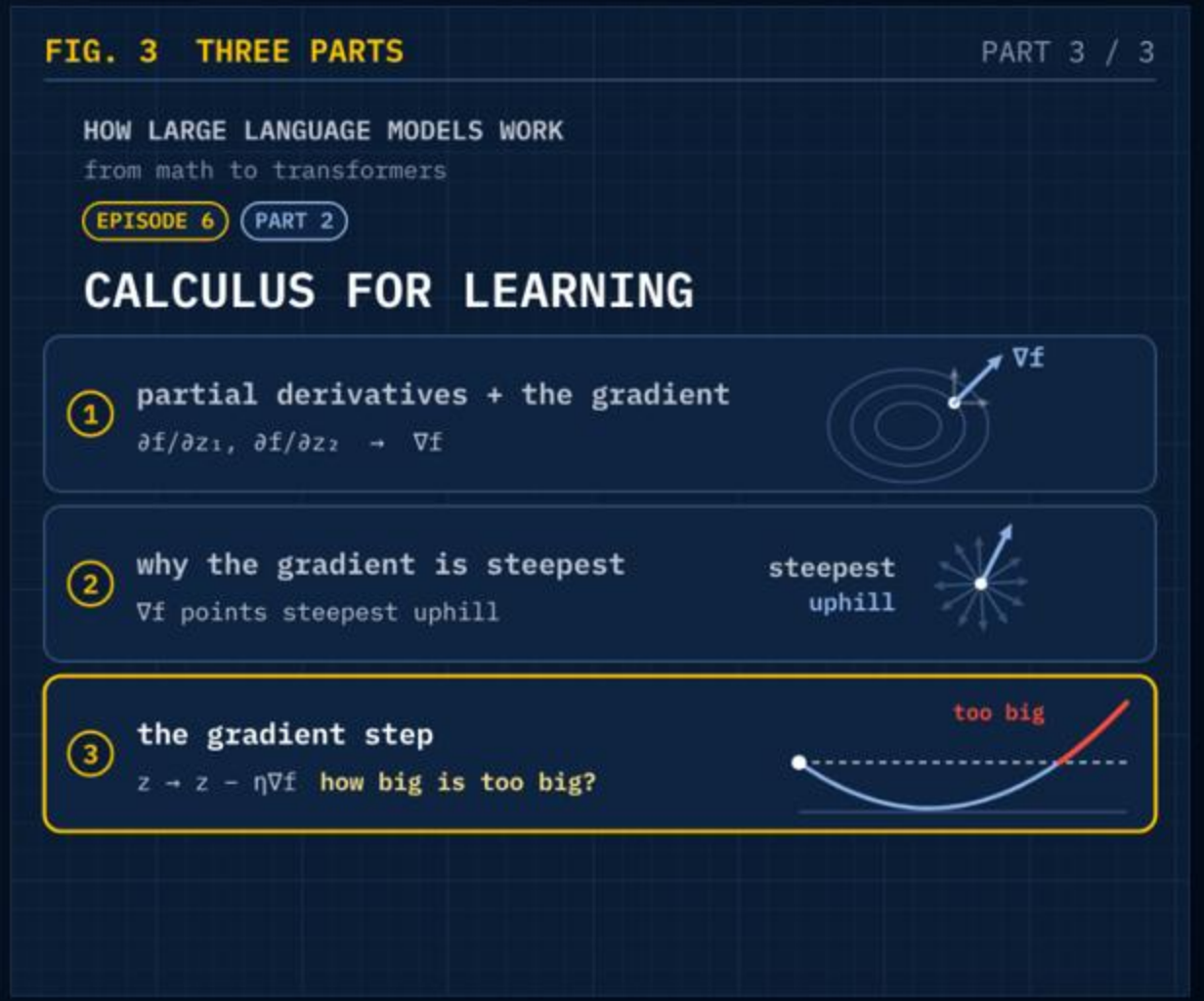
The gradient and the step

Three parts, every step worked by hand, and a notebook with every number.

01 Partial derivatives and the gradient

02 Why the gradient points steepest uphill

03 The gradient step, and how big is too big



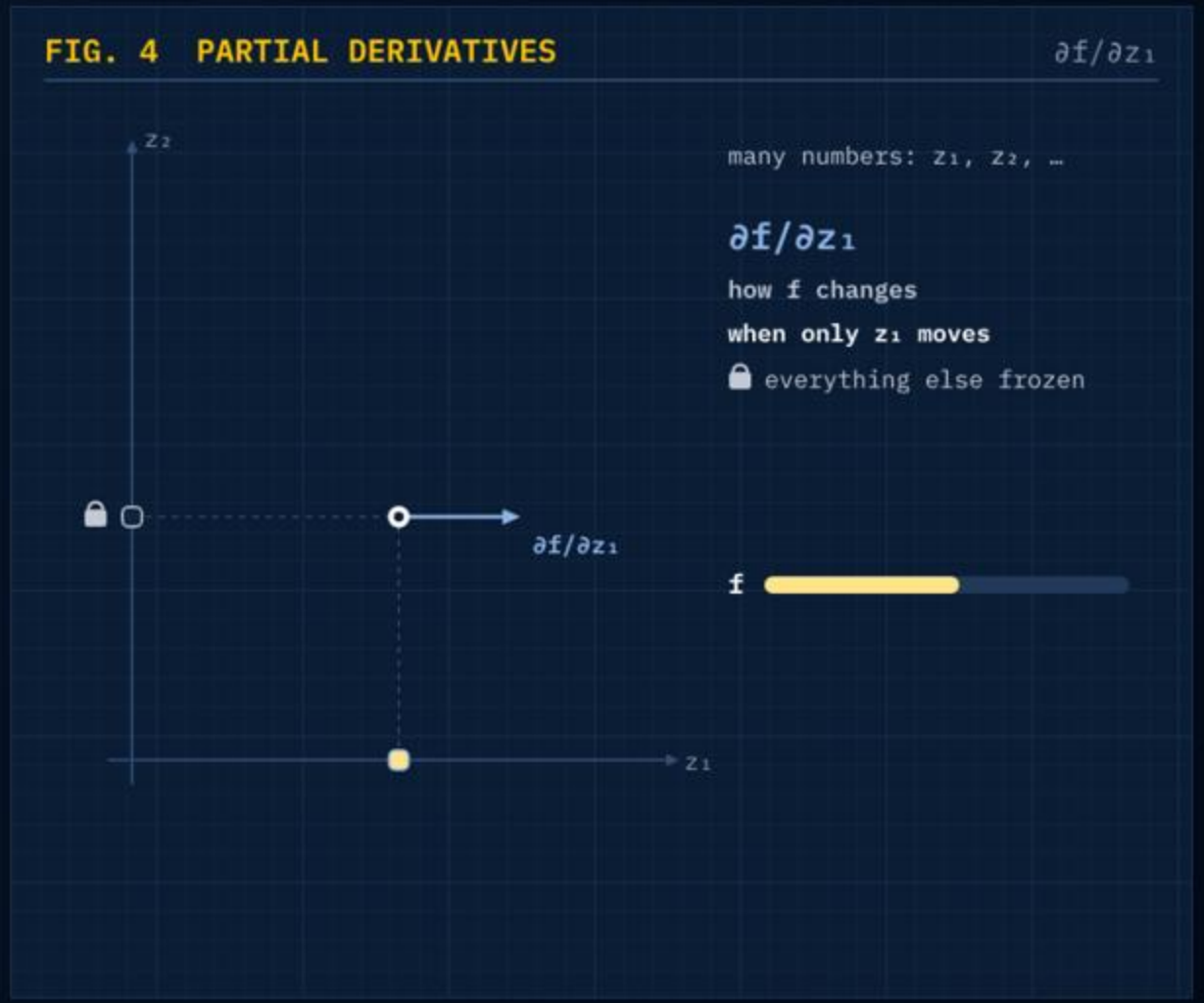
Companion notebook (Colab), linked in the description. Sources on every slide.

MARK

Many numbers, many slopes

$\partial f/\partial z_1$: change only z_1, freeze the rest.

01 One partial derivative per input



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 4 §4.3 (gradient-based optimization).

PART 1 · YOUR FIRST REAL GRADIENT

You can’t move one alone

Softmax keeps the total at 1, so the knob is the score.

01 mat −0.70 · floor +0.20 · sofa +0.15 · other +0.35



Worked example; every number is checked in the companion notebook. Episode 1’s probabilities (mat 0.30, floor 0.20, sofa 0.15, other 0.35) are illustrative, not measured.

PART 1 · A TEST LANDSCAPE

A stretched bowl

$$f(z_1, z_2) = z_1^2 + 2z_2^2$$

01 $\partial f/\partial z_1 = 2z_1$

02 $\partial f/\partial z_2 = 4z_2$

03 Contours: ellipses around (0, 0)

FIG. 6 THE TEST BOWL — TOP VIEW

$f(z_1, z_2) = z_1^2 + 2z_2^2$ seen from above

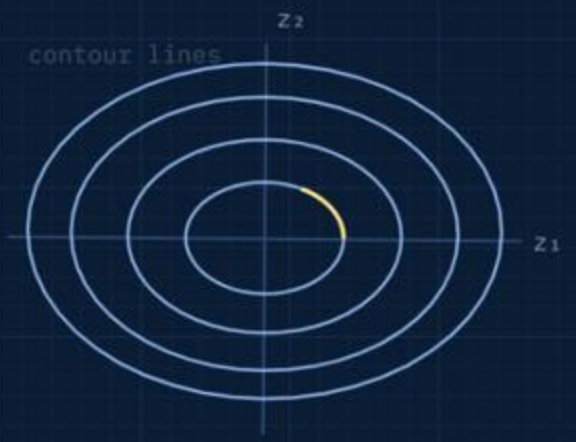
power rule: d/dz $z^2 = 2z$

$\partial f/\partial z_1 = 2z_1$

$\partial f/\partial z_2 = 4z_2$

$= 2 \cdot 2z_2 = 4z_2$

that 2 out front



The bowl $f = z_1^2 + 2z_2^2$ is a teaching example. Worked example; every number is checked in the companion notebook.

PART 1 · BY HAND

The gradient at (1, 1)

$f(1, 1) = 1 + 2 = 3$

01 $\partial f/\partial z_1 = 2 \cdot 1 = 2 \cdot \partial f/\partial z_2 = 4 \cdot 1 = 4$

02 $\nabla f = (2, 4)$

03 $\|\nabla f\| = \sqrt{(4 + 16)} = \sqrt{20} \approx 4.47$

FIG. 7 BY HAND: THE GRADIENT

STEP 04 / 04

stand at the point (1, 1)

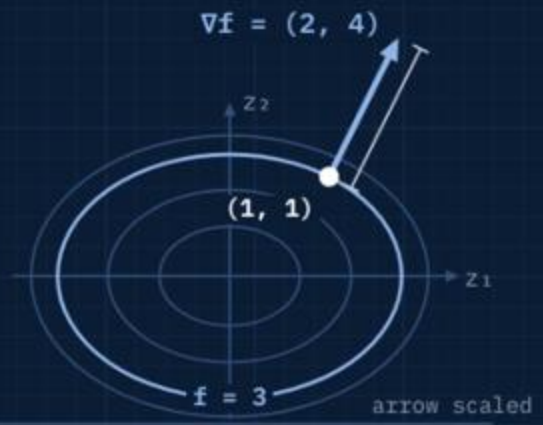
$f = 1^2 + 2 \cdot 1^2 = 1 + 2 = 3$

$\partial f/\partial z_1 = 2z_1 = 2 \cdot 1 = 2$

$\partial f/\partial z_2 = 4z_2 = 4 \cdot 1 = 4$

$\nabla f = (2, 4)$

how steep?



$\|\nabla f\| = \sqrt{(2^2 +}$

The bowl $f = z_1^2 + 2z_2^2$ is a teaching example. Worked example; every number is checked in the companion notebook.

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PART 2 · BY HAND FIRST

Test three directions

Slope in unit direction $a = \nabla f \cdot a$, with $\nabla f = (2, 4)$

01 $a = (1, 0)$: $2 \cdot 1 + 4 \cdot 0 = 2$

02 $a = (0, 1)$: 4

03 $a = (2, 4)/\sqrt{20}$: $20 \div 4.47 = 4.47$, steeper than both

FIG. 8 BY HAND: DIRECTIONS

DIRECTION 3 / 3

test it by hand

slope along a = ∇f · a

a: unit direction (length 1)

at (1, 1): ∇f = (2, 4)

go straight along z_1

a = (1, 0): 2·1 + 4·0 = 2

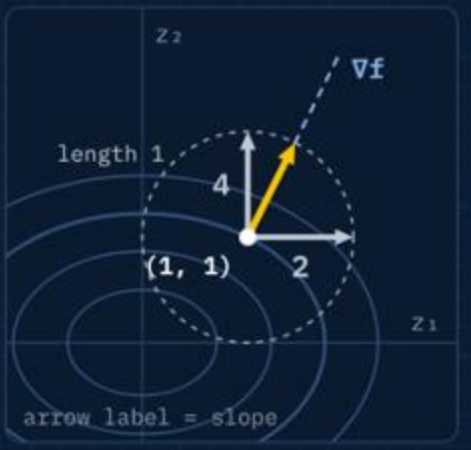
go straight along z_2

a = (0, 1): 2·0 + 4·1 = 4

along ∇f itself, shrunk to length 1

a = (2, 4) ÷ √20

√20 ≈ 4.47



slope along a



The bowl $f = z_1^2 + 2z_2^2$ is a teaching example. Worked example; every number is checked in the companion notebook.

The gradient points uphill

Slope in unit direction a: $\nabla f \cdot a = \|\nabla f\| \cos \varphi$

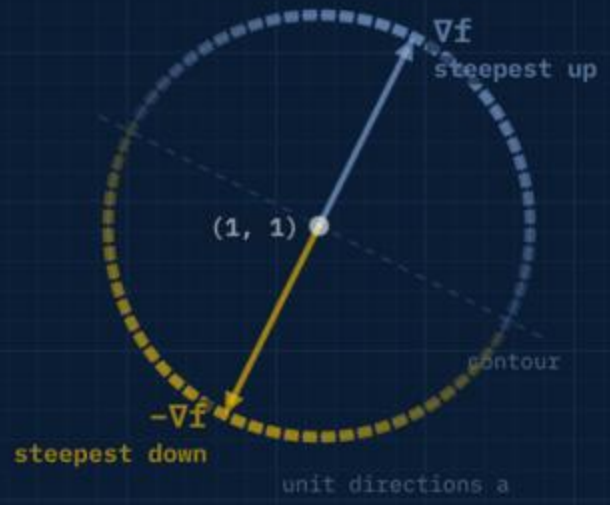
01 Biggest at $\varphi = 0$: ∇f is steepest up

02 $-\nabla f$ is steepest down

03 Fine print: smooth, $\nabla f \neq 0$, Euclidean distance

FIG. 9 WHY STEEPEST?

FINE PRINT



slope along unit a

$= \nabla f \cdot a$

$= \|\nabla f\| \, \|a\| \cos \varphi$

$\|a\| = 1$, so $\|\nabla f\| \cos \varphi$

three conditions

smooth (differentiable)

Guichard, *Calculus* §14.5, Directional Derivatives (LibreTexts). Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 4 §4.3 (gradient-based optimization).

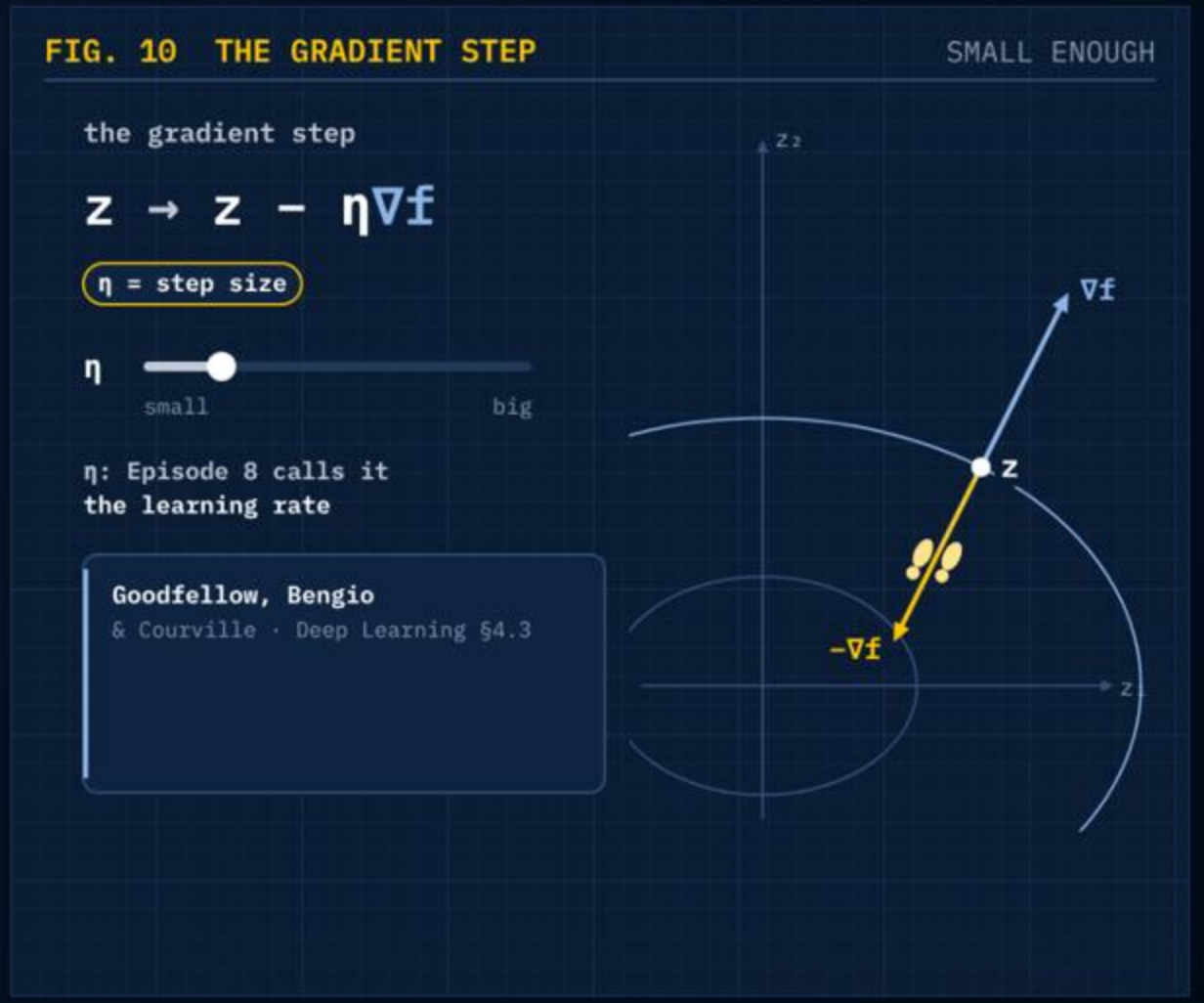
PART 3 · THE GRADIENT STEP

Step against the gradient

$z \to z - \eta \nabla f$

01 η: the step size (Episode 8: the learning rate)

02 Lowers f “for small enough” steps



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 4 §4.3 (gradient-based optimization).

PART 3 · BY HAND, H = 0.1

A small step

From (1, 1): $\eta\nabla f = (0.2, 0.4)$

01 New point (0.8, 0.6)

02 $f = 0.64 + 2 \times 0.36 = 0.64 + 0.72 = 1.36$

FIG. 11 A SMALL STEP — at (0.8, 0.6)

$\eta = 0.1$ from (1, 1)

$\eta\nabla f = 0.1 \times (2, 4)$

$= (0.2, 0.4)$

subtract

z_1: $1 - 0.2 = 0.8$

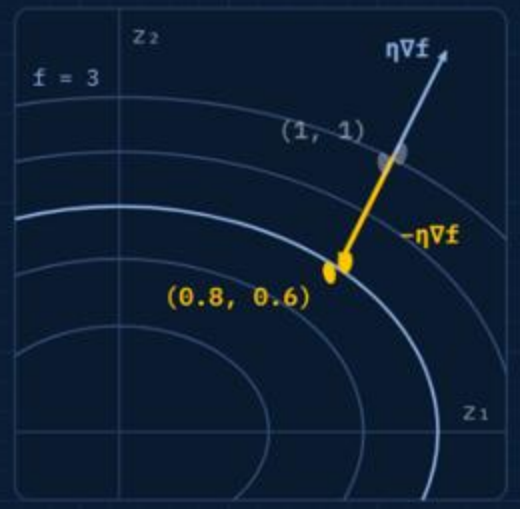
z_2: $1 - 0.4 = 0.6$

→ new point (0.8, 0.6)

new height $f = z_1^2 + 2z_2^2$

$0.8^2 = 0.64$

$2 \times 0.6^2 =$



height f

3

The bowl $f = z_1^2 + 2z_2^2$ is a teaching example. Worked example; every number is checked in the companion notebook.

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PART 3 · BY HAND, H = 1

A greedy step

$(1, 1) - (2, 4) = (-1, -3)$

01 $f = 1 + 2 \times 9 = 1 + 18 = 19$

FIG. 12 A GREEDY STEP

f: 3 → 19

$\eta = 1$ greedy

$(1, 1) - (2, 4)$

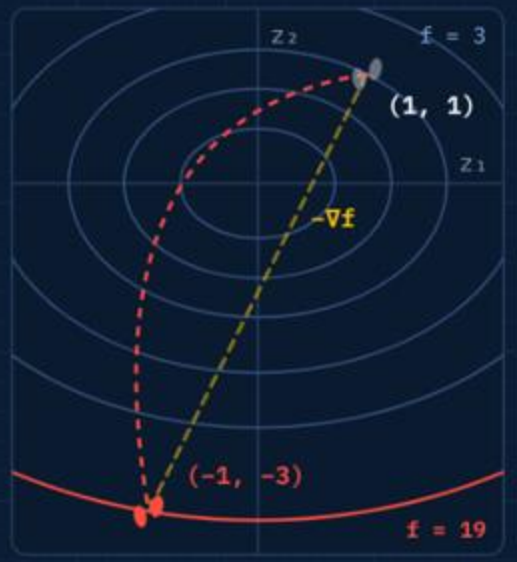
$= (-1, -3)$

height $f = z_1^2 + 2z_2^2$

$(-1)^2 = 1$

$2 \times (-3)^2 = 2 \times 9 = 18$

$f = 1 + 18 = \boxed{19}$



height f

3 19

The bowl $f = z_1^2 + 2z_2^2$ is a teaching example. Worked example; every number is checked in the companion notebook.

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PART 3 · WHY

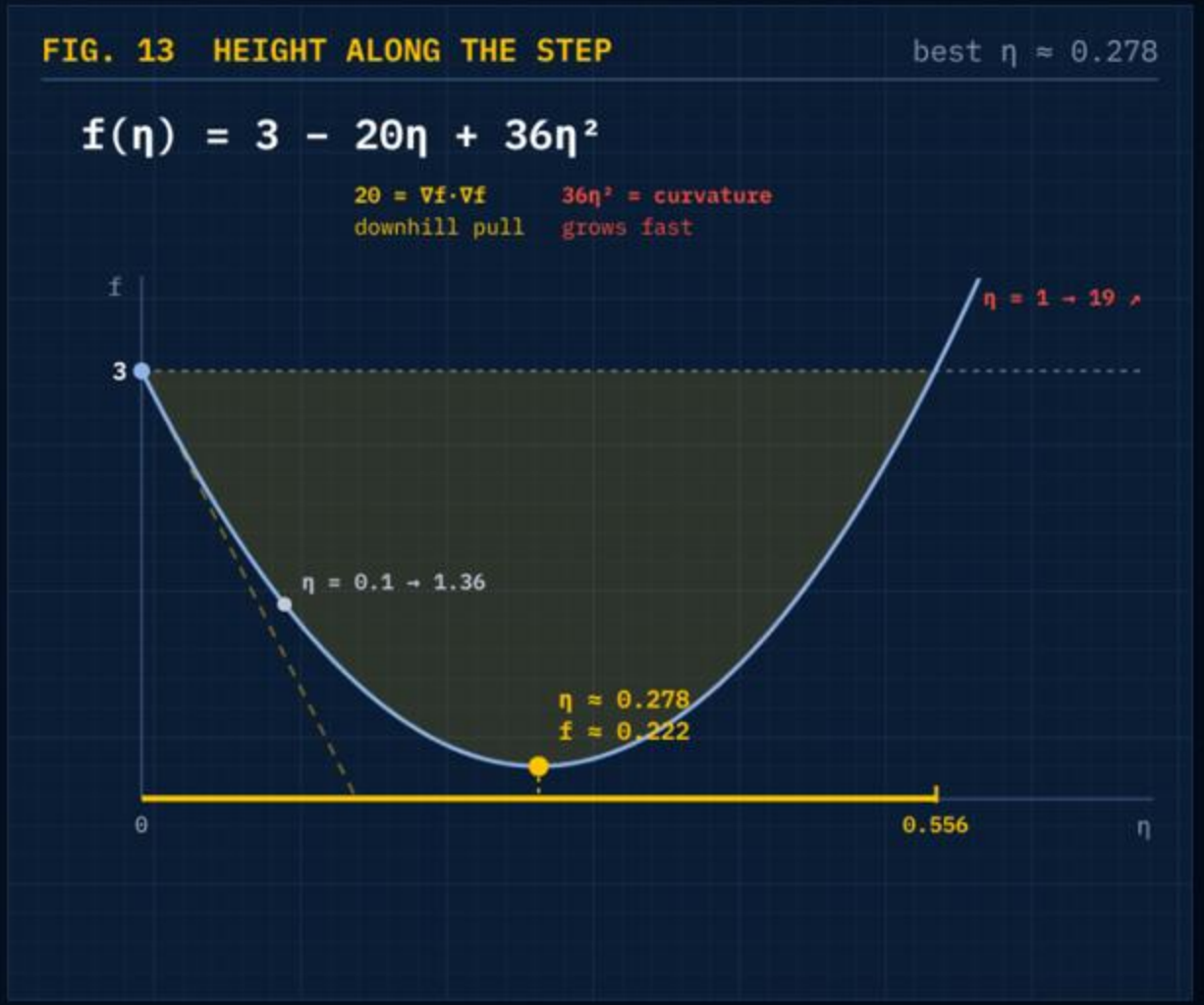
Curvature sets the limit

Along the line: $f(\eta) = 3 - 20\eta + 36\eta^2$

01 $20 = \nabla f \cdot \nabla f$: the downhill pull

02 $36\eta^2$: curvature, and it grows fast

03 Down only for $0 < \eta < 0.556$; best $\eta \approx 0.278$ → $f \approx 0.222$



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 4 §4.3.1 (curvature). The bowl $f = z_1^2 + 2z_2^2$ is a teaching example. Worked example; every number is checked in the companion notebook.

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PART 3 · DERIVE IT

Where the curve comes from

After a step η: (1 − 2η, 1 − 4η)

01 $(1-2\eta)^2 = 1-4\eta+4\eta^2$

02 $2(1-4\eta)^2 = 2-16\eta+32\eta^2$

03 Sum: $3-20\eta+36\eta^2$

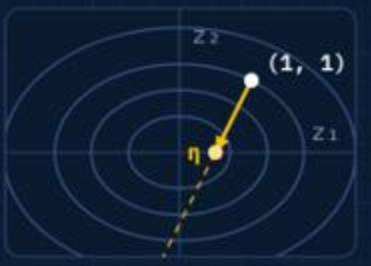
FIG. 14 DERIVE THE CURVE — STEP 4 / 4 ADD

goal: f(η) = 3 − 20η + 36η² — BY HAND

step η from (1, 1), ∇f = (2, 4)

point = (1 − 2η, 1 − 4η)

f = (1 − 2η)² + 2(1 − 4η)²



	$(1-2\eta)^2$	= 1	-4η	$+4\eta^2$
	$(1-4\eta)^2$	= 1	-8η	$+16\eta^2$
+	$2(1-4\eta)^2$	= 2	-16η	$+32\eta^2$
	$f(\eta)$	= 3	-20η	$+36\eta^2$

1 + 2 = 3

−4η − 16η = −20η

4η² + 32η² = 36η²

The bowl $f = z_1^2 + 2z_2^2$ is a teaching example. Worked example; every number is checked in the companion notebook.

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PART 3 · THE LIMITS, BY HAND

How big is too big

Back to 3 when $36\eta^2 = 20\eta$

01 $\eta = 20/36 = 0.556$: longer steps land higher

02 Best step: $20/72 \approx 0.278$

03 Height there: $3 - 5.56 + 2.78 \approx 0.222$

FIG. 15 THE LIMITS BY HAND

best $\eta \approx 0.278$

$f(\eta) = 3 - 20\eta + 36\eta^2$

back to 3 when

$3 - 20\eta + 36\eta^2 = 3$

$36\eta^2 = 20\eta$ (the 3s cancel)

$\div \eta$: $36\eta = 20$

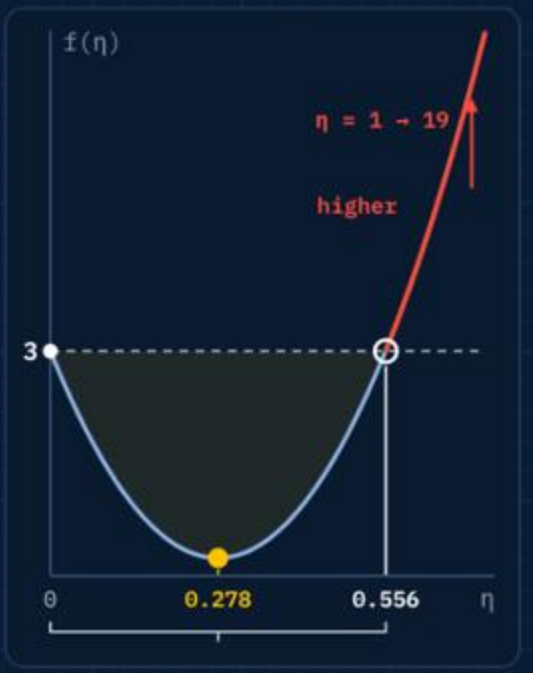
$\eta = 20/36 = 0.556$

$\eta > 0.556$: lands higher than 3

best step: halfway $(20/36) \div 2$

$\eta = 20/72 \approx 0.278$

$f = 3 - \underbrace{5.56}_{20 \cdot 0.278}$



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 4 §4.3.1 (curvature). The bowl $f = z_1^2 + 2z_2^2$ is a teaching example. Worked example; every number is checked in the companion notebook.

A PICTURE TO KEEP

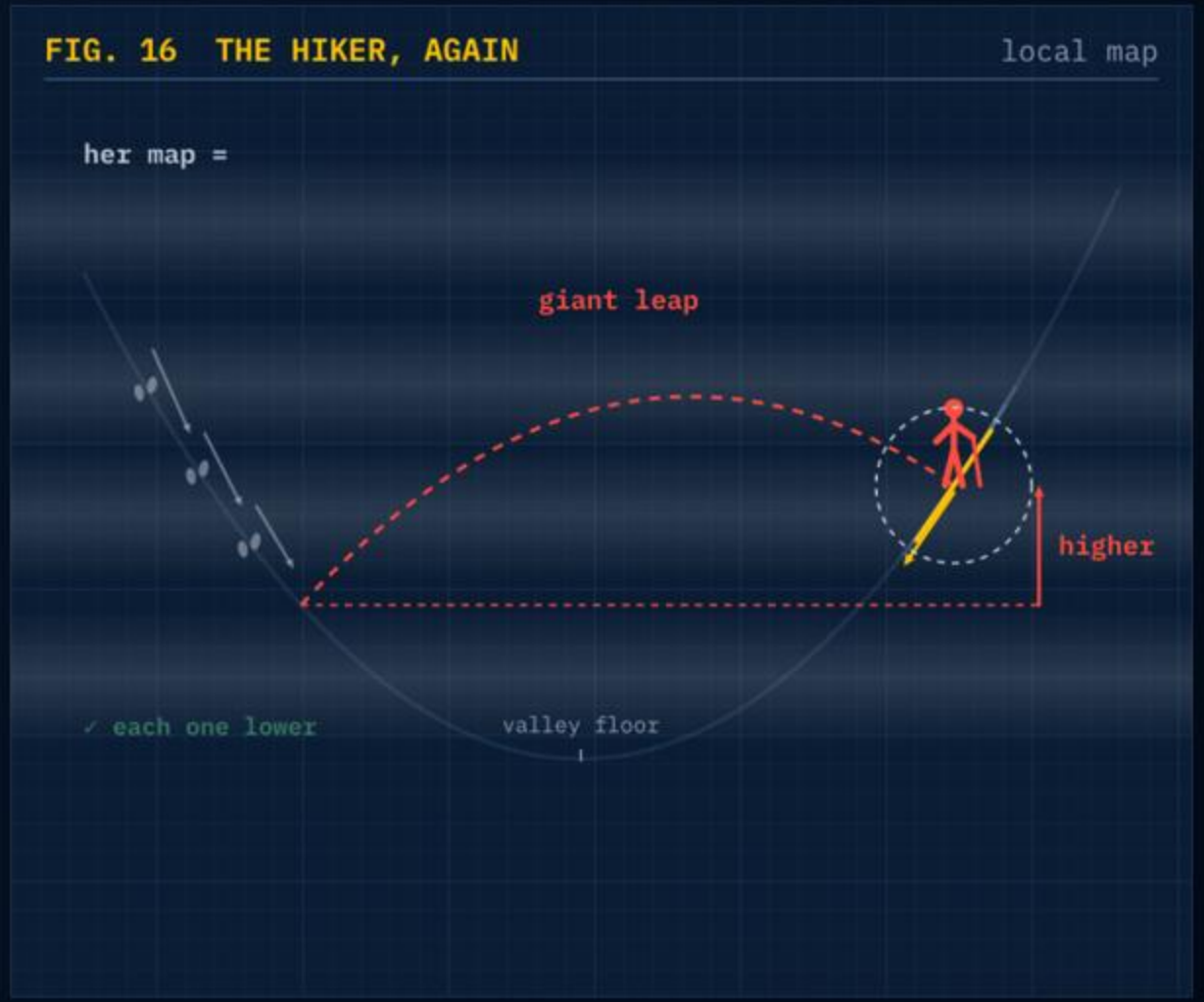
The hiker in the fog

Small steps go down. A giant leap overshoots.

01 Feel the tilt, step downhill

02 Leap too far: land higher on the far slope

03 Her map is only the ground under her boots



MYTH CHECK

“The gradient points to the bottom”

At (1, 1): $-\nabla f = (-2, -4)$; the bottom is along (−1, −1).

01 Dot product: $2 + 4 = 6$

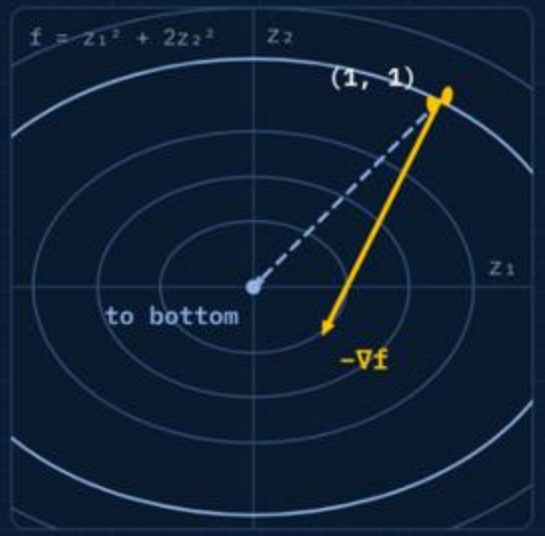
02 Lengths: $\sqrt{20} \times \sqrt{2} = \sqrt{40} \approx 6.325$

FIG. 17 MYTH CHECK

MYTH?

MYTH: the gradient points to the bottom

sounds right: near any point, −∇f does go down



$-\nabla f = (-2, -4)$

to bottom = (−1, −1)

by hand

dot: $(-2)(-1) + (-4)(-1)$

$= 2 + 4 = 6$

lengths: $\sqrt{20} \times \sqrt{2}$

$= \sqrt{40} \approx 6.325$

The bowl $f = z_1^2 + 2z_2^2$ is a teaching example. Worked example; every number is checked in the companion notebook.

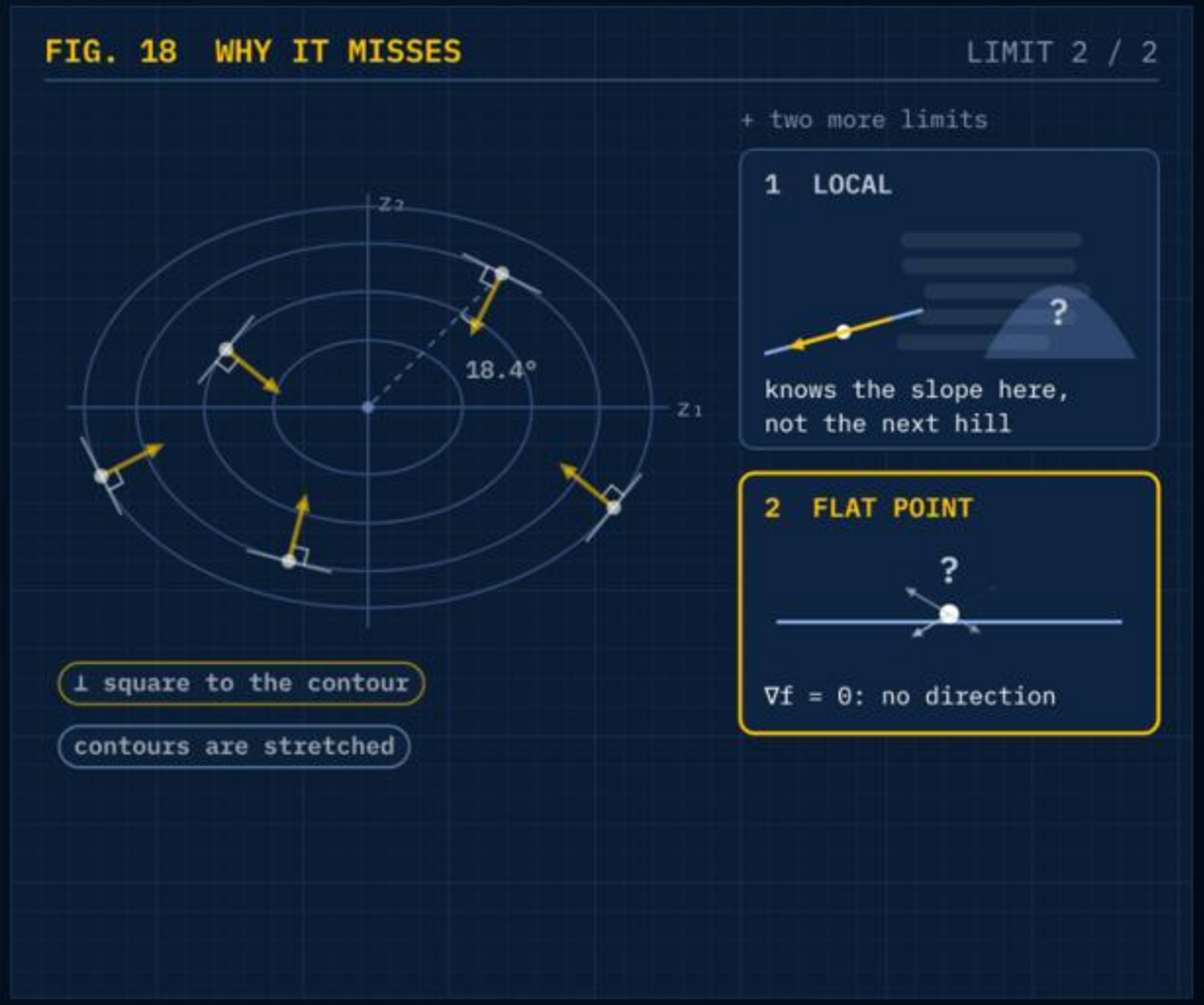
MYTH CHECK · WHY IT MISSES

Down from here ≠ the bottom

Steepest-down is square to the contour, and these contours are stretched.

01 The gradient is local

02 At a flat point it gives no direction



Goodfellow, Bengio & Courville, *Deep Learning* (MIT Press, 2016), ch. 4 §4.3 (gradient-based optimization). Guichard, *Calculus* §14.5, Directional Derivatives (LibreTexts).

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RECAP

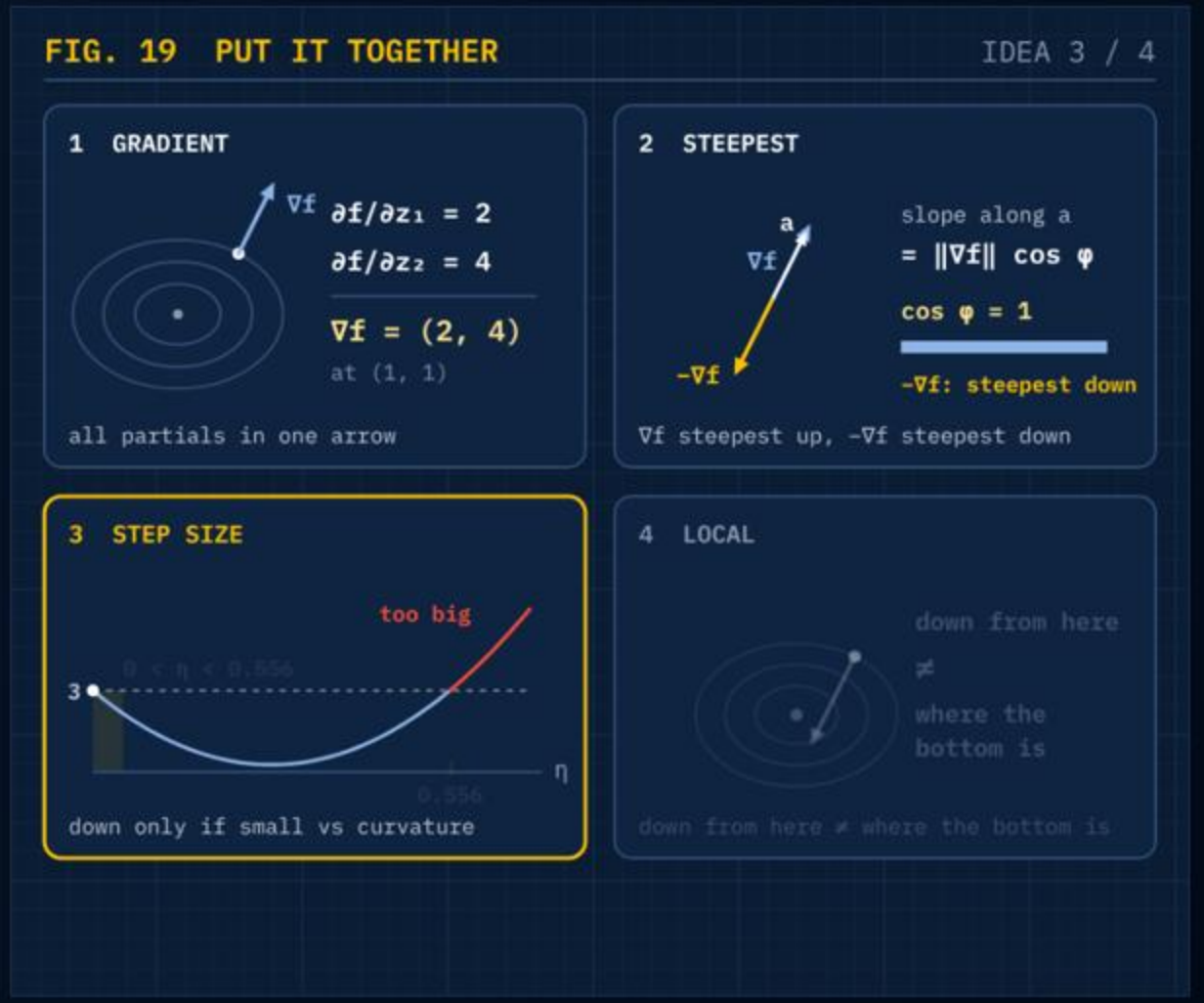
Four ideas

Every learning step in this series runs on these.

01 The gradient: every partial derivative in one arrow

02 It points steepest uphill; −∇f steepest down

03 A step goes down only if it's small vs the curvature



All sources are listed in the description.

MARK

YOUR TURN

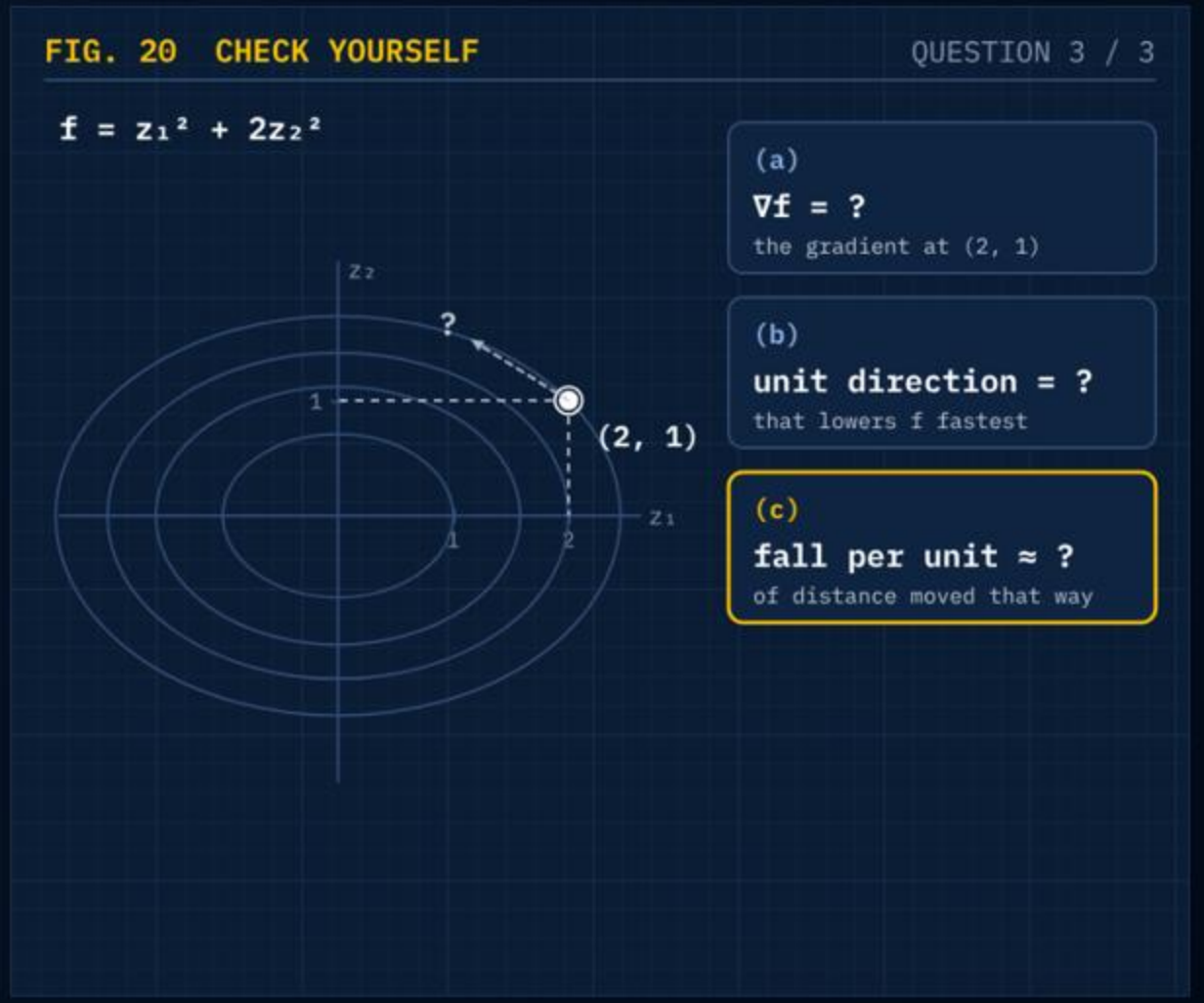
One question to take with you

Same bowl, $f = z_1^2 + 2z_2^2$, at the point (2, 1).

01 What is ∇f?

02 Which unit direction lowers f fastest?

03 How much does f fall per unit distance?



The bowl $f = z_1^2 + 2z_2^2$ is a teaching example. Worked example; every number is checked in the companion notebook.

YOUR TURN

The answer

Read the slope of a surface from its gradient.

01 $\nabla f = (2 \cdot 2, 4 \cdot 1) = (4, 4)$

02 Fastest down: $(-1, -1)/\sqrt{2}$

03 About $\sqrt{32} \approx 5.66$ per unit distance

FIG. 21 THE ANSWER

(a) the gradient

$$\nabla f = (2 \cdot 2, 4 \cdot 1) = \boxed{(4, 4)}$$

(b) fastest way down

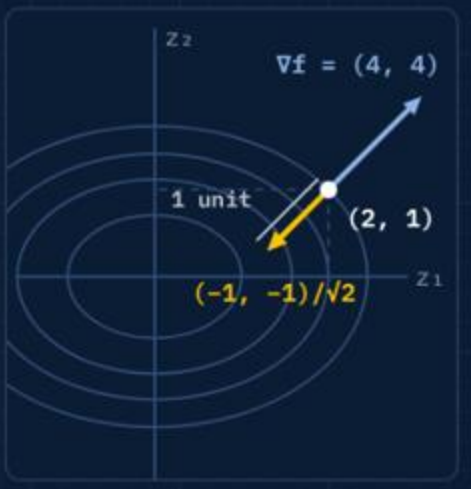
$$-(4, 4) \,/\, \|(4, 4)\| = \boxed{(-1, -1) \,/\, \sqrt{2}}$$

since $\|(4, 4)\| = 4\sqrt{2}$

(c) fall per unit distance

$$\|\nabla f\| = \sqrt{(4^2 + 4^2)} = \sqrt{(16 + 16)}$$

(b) unit direction



The bowl $f = z_1^2 + 2z_2^2$ is a teaching example. Worked example; every number is checked in the companion notebook.

TRY THIS WEEK

Three things to try.

01

From (1, 1), try η = 0.5 and η = 0.6 in the formula. Which side of 3 does each land on?

02

In the notebook, run gradient descent with your own step size. Watch it settle, or blow up

03

Find the slope in the direction (1, 1)/√2 by hand

Companion notebook and every source in the description.

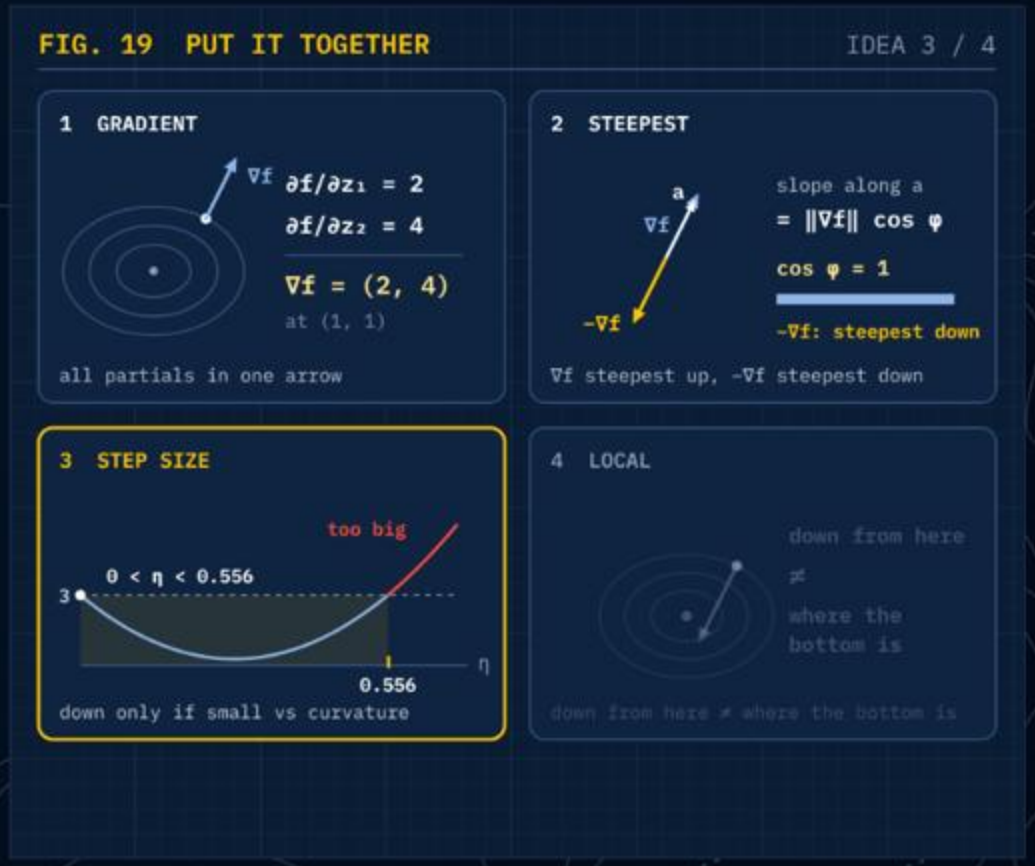
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NEXT TIME

Episode 7: What learning means. Thousands of surprisals. One number: the loss.

In the comments: which surprised you more, one step from 3 to 19, or the way down missing the bottom?

How Large Language Models Work · subscribe or follow for Episode 7



MARK